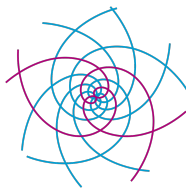


Stationary boundaries on the space of amenable subgroups

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$$C_r^*(G) = \overline{\text{span}\{\lambda_g : g \in G\}}^{\|\cdot\|} \subseteq \mathcal{B}(\ell^2(G)), \quad \lambda = \text{left reg. rep.}$$

- ▶ G is C^* -simple if $C_r^*(G)$ has no non-zero proper closed two-sided ideals.
- ▶ G has the **unique trace property** if $\tau(a) = \langle a\delta_e, \delta_e \rangle$ is the only tracial state on $C_r^*(G)$.
- ▶ The **amenable radical** $R_{\text{am}}(G)$ of G is its largest amenable normal subgroup.

Facts:

$$C^*\text{-simple} \implies R_{\text{am}}(G) = \{e_G\}.$$

$$\text{UTP} \implies R_{\text{am}}(G) = \{e_G\}.$$

Le Boudec '16: Non- C^* -simple groups with $R_{\text{am}}(G) = \{e_G\}$.

The *Furstenberg boundary* $\partial_F G$ is the universal G -boundary:
every compact, minimal, and strongly proximal G -space is a G -equivariant quotient of $\partial_F G$.

$$\text{Furman '03 : } \ker(G \curvearrowright \partial_F G) = \text{Rad}_{\text{am}}(G).$$

[Kalantar–Kennedy '17, Breuillard–Kalantar–Kennedy–Ozawa '17]

$$G \text{ is } C^*\text{-simple} \iff G \curvearrowright \partial_F G \text{ is free,}$$

$$G \text{ has the UTP} \iff \text{Rad}_{\text{am}}(G) = \{e_G\} \iff G \curvearrowright \partial_F G \text{ is faithful.}$$

- ▶ Chabauty space: $G \curvearrowright \text{Sub}(G)$ by conjugation.
- ▶ $\text{Sub}_{\text{am}}(G) = \{H \leq G : H \text{ is amenable}\}$ is closed and G -invariant.

Bader–Duchesne–Lécureux '16 + BKKO '17: G has a unique trace if and only if $\delta_{\{e_G\}}$ is the unique **invariant** measure on $\text{Sub}_{\text{am}}(G)$.

Hartman–Kalantar '23

G is C*-simple if and only if there is a non-degenerate $\mu \in \text{Prob}(G)$ for which $\delta_{\{e_G\}}$ is the unique μ -stationary measure on $\text{Sub}_{\text{am}}(G)$.

Many classes satisfy “for **every** μ ”, e.g. , free groups, linear groups, etc...

Question

Which (C*-simple) groups admit a non-trivial amenable SRS?

Hydration break



Fix $\mu \in \text{Prob}(G)$.

- ▶ A measure $\eta \in \text{Prob}(\text{Sub}(G))$ is μ -stationary if

$$\eta = \sum_{g \in G} \mu(g) g_* \eta.$$

Stationary measures always exist due to compactness (unlike invariant measures).

- ▶ Denote by $w_n = g_1 \cdots g_n$, $n \geq 1$, the right μ -random walk on G . Then the weak*-limit $\lim_{n \rightarrow \infty} (w_n)_* \eta$ exists a.s.

Definition: a μ -SRS η is a **boundary μ -SRS** if $\lim_{n \rightarrow \infty} (w_n)_* \eta$ is a.s. a Dirac mass.

Equivalently, if $(\text{Sub}_{\text{am}}(G), \eta)$ is a G -equivariant quotient of the Poisson boundary of (G, μ) .

The existence theorem

For finite $Q \subseteq G$ and $H \leq G$ define

$$\mathcal{U}_Q(H) = \{L \in \text{Sub}(G) : L \cap Q = H \cap Q\}.$$

Recurrent orbits property

Say that $H \neq \delta_{\{e_G\}}$ has *recurrent orbits* if for every finite $Q, Z \subseteq G$ there is $b \in G$ such that

$$(bz) \cdot H, (b^{-1}z) \cdot H \in \mathcal{U}_Q(H) \text{ for every } z \in Z.$$

Theorem [Cascioli – Gilabert Vio – S. '26]

Suppose $H \leq G$ is non-trivial and has recurrent orbits. Then there exists a symmetric $\mu \in \text{Prob}(G)$ with $\text{supp}(\mu) = G$ and finite entropy such that

$$w_n H w_n^{-1} \xrightarrow[n \rightarrow \infty]{\text{a.s.}} K \neq \delta_{\{e_G\}}.$$

If H is amenable, then $\eta = \text{Law}(K)$ is a non-trivial amenable boundary μ -SRS.

Let A, B be countable groups. Consider their wreath product

$$A \wr B := \bigoplus_B A \rtimes B.$$

Fact: If A is C^* -simple then so is $A \wr B$.

Corollary

Consider the group

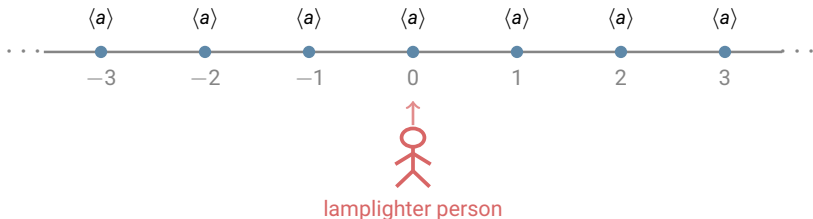
$$G = F_2 \wr \mathbb{Z}^3.$$

Then G is C^* -simple and yet **any** $\mu \in \text{Prob}(G)$ **with finite 1st moment (in particular, finite support) admits a non-trivial amenable boundary μ -SRS.**

I.e., the Hartman-Kalantar measures **must** have infinite first moment!

Start with $\langle a \rangle$ at every lamp

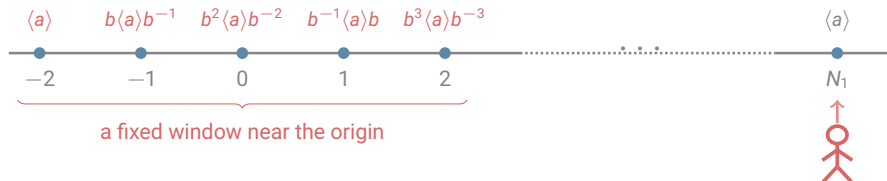
$$G = F_2 \wr \mathbb{Z}, \quad F_2 = \langle a, b \rangle, \quad H = \bigoplus_{k \in \mathbb{Z}} \langle a \rangle_k.$$



Every site carries the same cyclic subgroup, and the lamplighter starts at the origin.

Later: the lamplighter modifies lamps and moves

$$w_{n_1} = (\varphi_{n_1}, N_1), \quad N_1 \gg 1.$$

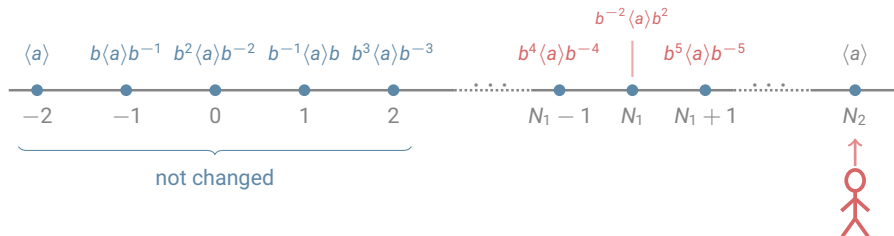


At the site k , the lamp value replaces $\langle a \rangle$ by

$$\varphi_{n_1}(k) \langle a \rangle \varphi_{n_1}(k)^{-1}.$$

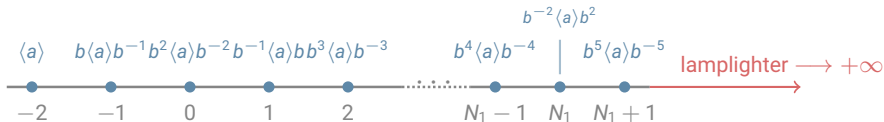
Much later: lamps near the origin stabilize

$$w_{n_2} = (\varphi_{n_2}, N_2), \quad N_2 \gg N_1 \gg 1.$$



already stabilized

possibly changed farther along the walk

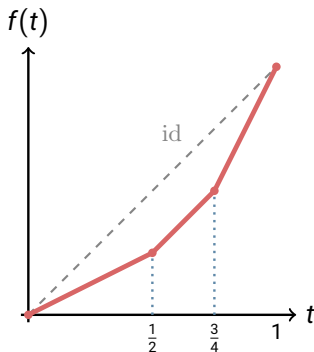


For every fixed $k \in \mathbb{Z}$, the value $\varphi_n(k)$ is eventually constant. Therefore

$$w_n H w_n^{-1} \longrightarrow H_\infty := \bigoplus_{k \in \mathbb{Z}} \varphi_\infty(k) \langle a \rangle \varphi_\infty(k)^{-1}.$$

Thompson's group F consists of the orientation-preserving, piecewise-linear homeomorphisms $f: [0, 1] \rightarrow [0, 1]$ with finitely many breakpoints, such that

$$\text{Break}(f) \subseteq \mathbb{Z}[1/2], \quad f'(t) \in 2^{\mathbb{Z}}.$$



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Corollary

There is a symmetric, non-degenerate, finite-entropy μ on Thompson's group F such that $\text{Sub}_{\text{ab}}(F)$ supports a non-atomic μ -boundary.

Matte Bon - Le Boudec '18: F is non-amenable if and only if it is C^* -simple.

Theorem [Cascioli – Gilabert Vio – S. '26]

Let η be a boundary SRS on $\text{Sub}(G)$ that is not a Dirac mass on a finite normal subgroup. Then η -almost every $H \leq G$ is *normalish*:

for every finite subset $Z \subseteq G$ the intersection $\bigcap_{z \in Z} zHz^{-1}$ is infinite.

Corollary

If G has no non-trivial finite normal subgroup and no amenable normalish subgroups, then for **every** non-degenerate μ the action $G \curvearrowright (B_\mu, \nu)$ is essentially free.

Applies to: groups with positive first ℓ^2 -Betti number, acylindrically hyperbolic groups, linear groups, etc... with trivial amenable radical.